A Hybrid Quantum–Classical Framework for Ground-State Correlation Analysis in Quantum Many-Body Systems

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ABSTRACT

Many-body quantum systems present analytical challenges owing to the frequent segregation of accurate benchmarking, efficient observable estimates, hardware validation, and predictive modelling. This study develops a cohesive hybrid quantum-classical framework for pairwise correlation analysis on a finite $2 \times 2 \times 2$ random-bond isotropic Heisenberg lattice, including 8-qubits and 12 nearest-neighbor interactions. The proposed methodology integrates sparse exact diagonalisation in PennyLane, Classical Shadow (CS) reconstruction from randomised local Pauli measurements, cross-platform validation in Qiskit, benchmarking on IBM hardware-aware noisy backends utilising the FakeFez backend with an ISA-transpiled Hadamard–CNOT proxy circuit, and supervised kernel-based learning of the coupling-to-correlation map derived from microscopic Hamiltonian parameters. The precise ground-state correlation matrix from PennyLane serves as the standard, whereas CS offers the nearest-approximation-based estimation, attaining a Root Mean Square Error (RMSE) of 0.06604 and a maximum deviation of 0.16267. However, the IBM FakeFez backend (IBM-FB) shows the largest discrepancies, reflecting the combined effects of proxy-state mismatch, backend-informed noise, and hardware-constrained transpilation. During the predictive phase, Gaussian-kernel regression (GKR) surpasses the neural-network-induced kernel surrogate (NNIKS), exhibiting a cross-validation RMSE of 0.06141 and a test RMSE of 0.11337. In general, the research provides a consistent framework for finite many-body quantum correlation analysis that brings together precise benchmarking, measurement-efficient reconstruction, examination of consistency across platforms, assessment of hardware-aware feasibility, and data-driven prediction.

Keywords: Quantum many-body systems, ground-state correlations, classical shadows, PennyLane, Qiskit Aer, IBM hardware-aware backend simulation.

Introduction

Many-body quantum systems are essential in numerous disciplines, including condensed matter physics, materials science, and quantum information theory. These systems frequently demonstrate intricate inter-particle interactions, rendering their precise modelling increasingly challenging as the system size expands. The Hilbert space expansion for these systems is exponential, posing considerable difficulties for computational techniques. Conventional classical methods often fail to deliver precise solutions for these systems due to the intrinsic complexity of quantum interactions in many-body physics[1].

In recent years, hybrid quantum-classical algorithms have attracted significant attention as potential solutions to these difficult problems. Using classical optimisation and machine learning (ML) techniques, these algorithms combine quantum computational power, which makes use of quantum phenomena such as superposition and entanglement, with the power of mathematics[2–4]. Specifically, quantum machine learning (QML) techniques have demonstrated their potential for extracting quantum correlations from quantum systems. These techniques offer a potential path to overcoming the restrictions associated with classical simulation techniques[5,6].

The extraction and analysis of quantum correlations, especially those associated with ground-state features, is a vital task in many-body quantum systems. Exact diagonalisation approaches, while precise, are computationally intensive and lack scalability for large systems. Conversely, techniques such as classical shadow tomography (CST), as presented in[7], offer a more efficient approach to estimating quantum observables while minimising measurement expenses. Classical shadows facilitate the effective estimation of diverse observables, such as spin correlation matrices, through the application of randomised measurements on a quantum state. These approaches have demonstrated effectiveness in reducing the resources required for state reconstruction, making them appropriate for noisy intermediate-scale quantum (NISQ) devices[8]. Furthermore, the CST has been demonstrated to be an efficacious technique for efficient state reconstruction, employing randomised Pauli measurements. This method significantly reduces the number of measurements required to obtain significant observables from quantum systems[9]. Classical shadows can be integrated with ML methodologies, such as GKR, to elucidate the connections between the microscopic Hamiltonian parameters of quantum states and their macroscopic correlation structures.

In this research work, we examine the resolution of a 3D simple cubic lattice characterised by the Heisenberg Hamiltonian and calculate its ground state correlation features utilising PennyLane and Qiskit Aer. We also provide a classical shadow formalism, an effective tomography technique that creates a classical representation of the quantum state, facilitating subsequent research to estimate correlation matrices[10]. The CST substantially reduces resource requirements and provides a scalable approach for computing many-body observables. The constructed state is used to train ML models, such as GKR and NNIKS, for regression to discern and forecast correlation patterns within the quantum system. This methodology emphasizes the synergy between quantum data generation and classical learning algorithms. The system is further simulated with the FakeFez IBM hardware-aware backend simulator to validate and compare the results. The outcomes from the simulated system were compared with those from IBM-FB hardware to validate the precision of our system's correlation matrices. The study's primary contributions are listed below.

- We propose a unified hybrid quantum–classical framework for an 8-qubit $2 \times 2 \times 2$ random-bond isotropic Heisenberg lattice, integrating exact diagonalization, classical shadows, cross-platform validation, and kernel-based learning.
- We demonstrate measurement-efficient correlation restoration, with classical shadows providing the closest practical approximation to the PennyLane benchmark among the non-exact methods.
- We also investigate a hierarchical validation strategy in which Qiskit provides software-level consistency, while the IBM FakeFez backend serves as a proxy-circuit hardware benchmark for quantifying the gap between ideal simulation and real-device execution.
- Finally, our approach showed that the coupling-to-correlation map is learnable from Hamiltonian parameters, with the Gaussian kernel Regression model outperforming the neural-network-induced kernel surrogate for predictive correlation modeling.

The study commences with the Introduction, which delineates the research challenge and introduces the suggested hybrid quantum-classical framework for analysing many-body ground-state correlations. Section 2 (Related Works) then examines pertinent literature on classical, quantum, and hybrid quantum-classical methodologies. Section 3 (Methodology) delineates the proposed framework, encompassing the physical system, Heisenberg model, lattice structure, coupling generation, precise ground-state computation, classical shadow reconstruction, cross-platform validation utilising PennyLane, Qiskit, and IBM-FB, ML dataset formulation, kernel-based models, and evaluation metrics. Section 4 (Result Analysis) presents the experimental evaluation and encompasses a correlation-matrix comparison. Section 5 (Conclusions) succinctly encapsulates the principal findings, constraints, and prospective avenues for further research.

Related Works

This section provides a literature survey on many-body quantum systems with a focus on the difficulties of ground-state correlation modelling. Yanming et al.[11] sought to improve ML for forecasting quantum many-body states and attributes by lowering sample complexity from exponential to polynomial. The method was numerically tested on a 1D quantum XY model with $n = 5$ qubits, uniform samples, and 30 independent runs. It used theoretical training sets of quantum states or state characteristics. In particular, the rectangular Fejér kernel with $\Lambda = 50$ served as the primary model's positive-excellent kernel estimator. With a ground state energy $R^2 \approx 0.99$ and a spin-spin correlation $R^2 \approx 0.98$, the numerical tests demonstrated strong performance. The goal of Laura et al.[12] geometric locality-based inductive bias was to provide improved ML predictions of quantum many-body ground state features. The approach was evaluated on 2D random Heisenberg models with spins ranging from 4×5 to 9×5. By utilizing random Fourier features in trials, the model implemented a nonlinear geometric feature map with ℓ_1-regularized LASSO regression. With a training and prediction time of $O(nN)$ and a sample complexity of $N = O(\log n)$ constant error, it outperformed the Dirichlet kernel, Gaussian kernel, and baselines of infinite-width neural networks in terms of RMSE. In another study, Willem et al.[13] used reinforcement learning (RL) to determine the ground states of quantum many-body lattice systems. Experiments were conducted on (6×6) and (10×10) Ising lattices to test stoquastic spin models, primarily the transverse field Ising and XXZ models. Using Soft Q-learning and a convolutional neural network (CNN), the authors applied three RL formulations based on the Feynman-Kac formula and the Schrödinger equation. According to the (6×6) Ising model, the RL methods had ground-state energy errors of 0.09%, 0.05%, and 0.04%. They also increased sampling speed by about 6× to 10× in settings with multiple spin flips. By learning quantum restrictions among expectation values rather than solving the entire ground state, Pei-Lin et al.[14] aimed to predict quantum many-body ground-state features. It utilised 1.92×10^6 physical fermion samples with 7.68×10^6 contrasting samples, and 6.75×10^4 spin-chain samples paired with 2.025×10^5 contrasting samples. The approach used supervised artificial neural networks to learn constraint penalty functions, followed by traditional constrained energy minimization. The findings strongly aligned with Hartree-Fock, theoretical, and DMRG benchmarks, exhibiting minimal constraint penalties ranging from 10^{-3} to 10^{-6}. Additionally, Jiang et al.[15] developed quantum algorithms on near-term qubit devices with nearest-neighbor couplings to simulate strongly correlated fermionic systems, particularly the 2D Fermi-Hubbard model. It looked at computational resource scaling for N fermionic modes or qubits instead of using a dataset. The proposed approaches comprised the following: Trotterized Hubbard model simulation, fermionic Gaussian state preparation, Slater determinant preparation, and $2D$ fermionic Fourier transformation (FFT).

In order to reconstruct quantum many-body states, Giacomo et al.[16] conducted experiments using a Rydberg atom quantum simulator and noisy measurements. Using 15 time steps and around 3,000 samples per step, it mainly utilised bit string data from $N = 8$ trapped ^{87}Rb atoms. The technique used a noise-layered restricted Boltzmann machine (RBM) wavefunction to fix measurement mistakes. The results demonstrated better reconstruction: fidelities exceeded 90%, and the one-body, two-body, and Rényi mutual information observables were accurately predicted. Nomura et al.[17] examined neural network-based quantum many-body solutions for strongly coupled electron systems. It concentrated on RBM in conjunction with pair-product wave functions, integrated with VMC, stochastic reconfiguration, Lanczos enhancement, and MACE. Notable examples encompassed the Hubbard model, $(J_1) - (J_2)$ Heisenberg model, dmit salts, and cuprates. The primary findings encompassed enhanced energy precision, a QSL phase within the range $0.49 \lesssim \frac{J_2}{J_1} \lesssim 0.54$, accurate reproduction of the dmit phase, and cuprate trends aligning with $T_c^{\mathrm{opt}} = 0.16|t_1|F_{\mathrm{SC}}$. With sample complexity independent of system size, Marc et al.[18] investigated the estimation of ground-state features of n-qubit gapped local Hamiltonians. It used training data generated by DMRG and matrix product state (MPS) and $N = 409$ to 3686 samples to evaluate 2D random Heisenberg models with 20 to 45 qubits. The methods employed were modified ridge regression and a deep neural network (DNN) with local neural models trained using AdamW. The experimental results demonstrated that the deep learning model achieved a stable prediction error as the system size increased and an RMSE approximately half that of the previous regression method. Subsequently, Mehta et al.[19] examined the potential of QML in facilitating many-body physics simulations and quantum state reconstruction. It examined hybrid quantum-classical methods such as Variational Quantum Eigensolver (VQE), Quantum Neural Network (QNN), Quantum Approximate Optimization Algorithm (QAOA), Quantum Generative Adversarial Network (QGAN), and Quantum Convolutional Neural Network (QCNN). Metrics like mean absolute error, energy variance, and fidelity were highlighted in the study. It was found that QML can enhance phase recognition, ground-state estimation, and state reconstruction; however, hardware noise, encoding issues, barren plateaus, and scalability remain significant constraints. Chen et al.[20] introduced Robust Shadow Estimation (RShadow), a noise-resistant adaptation of traditional shadow tomography for assessing quantum state characteristics on noisy NISQ devices. It incorporates a calibration phase to identify and rectify gate and measurement noise via classical postprocessing. The approach operates with both global and local Clifford measurements while maintaining near-standard shadow sample efficiency in the presence of bounded noise. Numerical evaluations of GHZ fidelity, TFIM correlations and energy, and (H_2) ground state energy indicate that RShadow maintains accuracy across various noise models, but the standard shadow estimate exhibits bias.

Tran et al.[21] investigated the efficiency of CS and graph neural networks in predicting the ground-state features of quantum Hamiltonians. The authors utilised $2D$ antiferromagnetic Heisenberg models to produce ground-state data using DMRG and simulated $T = 10^3$ Pauli measurements to generate classical shadow labels. They conducted a comparison of GNNs, MLPs, and NTK models utilising between 10 and 320 training Hamiltonians. The findings indicate that GNNs, particularly the pairwise GNN, achieved optimal sample efficiency, demonstrating the utility of Hamiltonian graph structures for learning many-body ground-state properties. In another study, Rahim et al.[22] introduced CD-QAOA, a reinforcement learning-optimized version of QAOA, influenced by counterdiabatic driving for rapid many-body ground-state preparation. The approach integrates continuous optimisation of unitary durations with reinforcement learning-based selection of operator sequences utilising adiabatic gauge potential terms. Experiments on spin $\frac{1}{2}$ and spin 1 Ising models, the spin 1 Heisenberg chain, and the Lipkin Meshkov Glick model demonstrate that CD-QAOA attains reduced energy and enhanced fidelity compared to normal QAOA, particularly for brief procedures. It also delineates a quantum speed limit of around $T_{\mathrm{QSL}} \approx 4.5$ and demonstrates that protocols acquired from tiny systems may be generalised to bigger systems. Struchalin et al.[9] empirically demonstrated the CST for assessing quantum-state characteristics without complete state reconstruction. The authors estimated observable expectation values and state integrity using random stabiliser observations on high-dimensional photon spatial states with dimensions $D = 2, 4, 8, 16,$

and 32. The approach demonstrated robust concordance with direct measurements, exhibiting Pearson correlations ranging from 0.989 at $D = 2$ to 0.875 at $D = 32$, utilising 10^4 stabiliser measurements and 5000 random projectors. The findings indicate that CS provides unbiased estimates despite missing observations, whereas maximum likelihood estimation yields biased results.

Methodology

Qubit

A quantum bit, or qubit, is the fundamental building block of quantum computing. Although classical bits may only exist in one of two states—0 or 1—qubits, which are quantum mechanical systems with two levels, can exist in a linear superposition of both base states at the same time. A qubit is formally defined as a normalised state vector within a two-dimensional complex Hilbert space $\mathscr{H}_2$[23]. It can be expressed as its general pure state:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \tag{1}$$

where the orthonormal basis vectors $|0\rangle$ and $|1\rangle$ are computational basis states, the complex probability amplitudes $\alpha, \beta \in \mathbb{C}$, and the total probability conservation is guaranteed by the normalisation condition.

$$|\alpha|^2 + |\beta|^2 = 1 \tag{2}$$

In the case of a computational basis measurement, the qubit collapses probabilistically to $|0\rangle$ with a probability of $|\alpha|^2$ or to $|1\rangle$ with a probability of $|\beta|^2$.

Fundamental Quantum Gates: The Building Blocks of Quantum Computation

Quantum computers rely on unitary operations called quantum gates to interact with qubits. They maintain probability while controlling superposition, phase, and entanglement. While multi-qubit gates establish interactions between qubits, single-qubit gates alter the states of a single qubit. When put together, they constitute the fundamental elements of quantum algorithms[24] and circuits[25]. Table 1 presents various types of elementary quantum gates.

Number of Qubits	Gate Name	Matrix Form	Description
Single-Qubit	Identity Gate	$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	Leaves the qubit unchanged.
	Pauli X	$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	Performs a bit flip.
	Pauli Z	$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	Performs a phase flip.
	Pauli Y	$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$	Combines X and Z with phase factor.
	Hadamard Gate	$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$	Creates equal superposition states.
	Phase Gates	$S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}, \quad T = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$	Modifies only the relative phase.
	Rotation Gates	$R_x(\theta) = e^{-i\theta X/2}$, $R_y(\theta) = e^{-i\theta Y/2}, ; R_z(\theta) = e^{-i\theta Z/2}$	Continuous Bloch sphere rotations.
Two-Qubit	Controlled-NOT (CNOT)	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$	Flips target if control is $\|1\rangle$.
	SWAP Gate	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	Exchanges two qubits.
Multi-Qubit	Toffoli Gate (CCNOT)	$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$	Flips target if both controls are $\|1\rangle$.

Table 1. Fundamental quantum gates.

Quantum Circuit

This study utilises quantum circuits as the operational foundation for executing unitary transformations on qubit registers. A quantum circuit consists of a series of quantum gates applied to an initial state, usually $|0\rangle^{\otimes N}$, facilitating the creation of parameterised quantum states and the assessment of observables.

As part of the suggested architecture, quantum circuits are employed in different contexts: (i) reference correlation computation through precise state-vector simulation in PennyLane, (ii) noise-aware simulation through transpiled circuit execution in Qiskit, and (iii) experimental validation under realistic noise conditions through hardware execution on IBM quantum devices.

Quantum Many-Body System and Hamiltonian Formulation

The quantum many-body spin system we'll look at is based on a three-dimensional (3D) cube lattice made up of $N = N_x N_y N_z$ spin-$\frac{1}{2}$ qubits. Each lattice site represents a quantum degree of freedom, and interactions are limited to nearest-neighbor pairings established by the lattice connection. The system is represented through a disordered isotropic Heisenberg Hamiltonian, effectively encapsulating the exchange interactions among neighbouring spins while integrating randomness in the coupling strengths. The Hamiltonian is defined as:

$$H = \sum_{(i,j)\in E} J_{ij}(X_i X_j + Y_i Y_j + Z_i Z_j) \tag{3}$$

where X_i, Y_i, Z_i are Pauli operators acting on qubit i, and E represents the collection of nearest-neighbor edges in the lattice graph. By individually sampling the coupling coefficients J_{ij} from a uniform distribution $J_{ij} \sim \mathscr{U}(0,1)$, quenched disorder is introduced into the system, resulting in an ensemble of Hamiltonians with different interaction strengths but the same lattice structure.

Ground-State Computation and Reference Quantum State

The ground state $|\psi_0\rangle$ for every Hamiltonian realisation is found by resolving the eigenvalue problem:

$$H|\psi_0\rangle = E_0|\psi_0\rangle \tag{4}$$

where the Hamiltonian's lowest eigenvalue is denoted by E_0. The ground state is calculated using sparse iterative diagonalisation techniques because the Hamiltonian resulting from local interactions is sparse. The resultant ground state functions as the precise quantum reference state for all ensuing investigations, encompassing observable computation, supervised learning, and fidelity assessment.

Physical Observable: Spin Correlation Matrix

The spin-correlation matrix $\mathbf{C} \in \mathbb{R}^{N\times N}$ is the main observable examined in this paper. It represents pairwise quantum correlations throughout the lattice. Every entry has a definition of

$$C_{ij} = \frac{1}{3}\left(\langle X_i X_j\rangle + \langle Y_i Y_j\rangle + \langle Z_i Z_j\rangle\right), \tag{5}$$

where the ground state $|\psi_0\rangle$ is used to calculate the expectation values. For consistent self-correlation scaling, the diagonal elements are normalised so that $C_{ii} = 1$. The desired quantity for both classical shadow reconstruction and ML prediction challenges is this matrix, which gives a compact representation of the emerging many-body structure.

Classical Shadow Estimation

The classical shadow approach is used to allow scalable estimation of high-dimensional quantum observables. At each measurement round, in this framework, a randomly selected Pauli basis is used to independently measure each qubit. The measurement outcome for each qubit i and snapshot t is recorded as $m_i^{(t)} \in \{-1,+1\}$, which represents the eigenvalue of the measured Pauli operator, and a measurement basis $b_i^{(t)} \in \{X,Y,Z\}$ is chosen uniformly at random. By carrying out this process for T separate snapshots, the dataset is obtained.

$$\mathscr{S} = \left\{\left(\mathbf{b}^{(t)}, \mathbf{m}^{(t)}\right)\right\}_{t=1}^{T}, \tag{6}$$

where $\mathbf{b}^{(t)}$ represents the vector of measurement bases and $\mathbf{m}^{(t)}$ represents the vector of outcomes for the t-th snapshot, respectively. With this method of measurement, the number of observations needed to efficiently estimate an exponentially large number of observables grows linearly with the size of the system, rather than increasing exponentially.

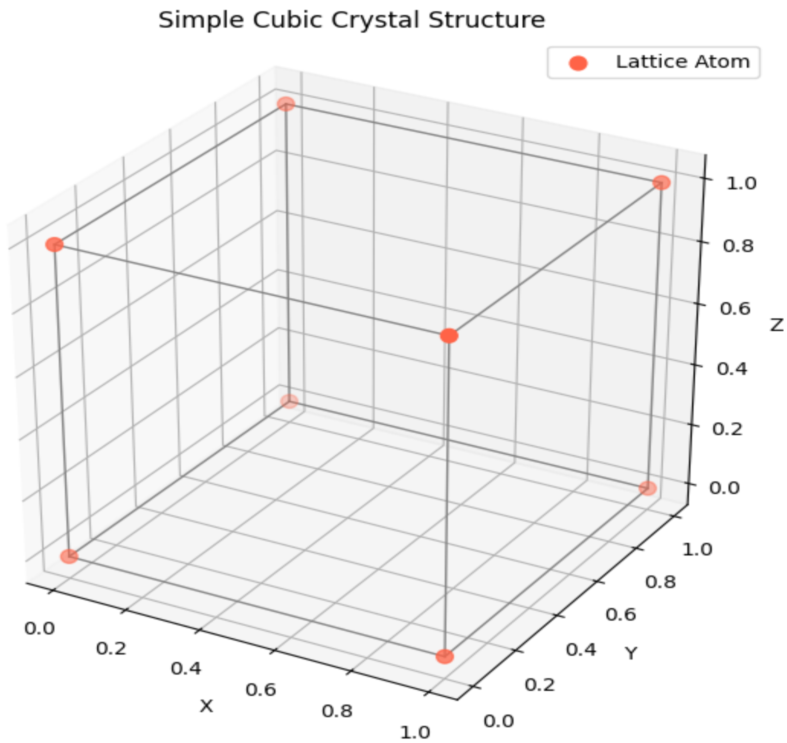

Figure 1. Finite $2 \times 2 \times 2$ cubic lattice with nearest-neighbor qubit interactions.

Shadow State Reconstruction

A local classical shadow estimator is generated from each measurement snapshot using the inverse measurement channel. An estimator is defined for each qubit i as

$$\hat{\rho}_i^{(t)} = 3U^{\dagger}_{b_i^{(t)}} \left| s_i^{(t)} \right\rangle \left\langle s_i^{(t)} \right| U_{b_i^{(t)}} - I, \tag{7}$$

Here, $U_{b_i^{(t)}}$ signifies the unitary rotation correlated to the measurement basis $b_i^{(t)}$, whereas $\left| s_i^{(t)} \right\rangle$ denotes the post-measurement eigenstate connected with the result $m_i^{(t)}$.

By taking the tensor product of all qubits, we may obtain the estimator for the entire system that is mention in Eq 8.

$$\hat{\rho}^{(t)} = \bigotimes_{i=1}^{N} \hat{\rho}_i^{(t)} \tag{8}$$

The final optimum classical-shadow reconstruction is calculated by averaging over all T measurement snapshots.

$$\hat{\rho} = \frac{1}{T} \sum_{t=1}^{T} \hat{\rho}^{(t)} \tag{9}$$

This estimator offers an efficient approximation of the quantum state without requiring complete quantum state tomography.

Observable Estimation and Correlation Reconstruction

The classical-shadow dataset is used to directly estimate the expectation values of Pauli-string observables given measurement snapshots that are compatible with the observable of interest. Using a median-of-means estimator strengthens robustness when sampling is limited. In order to get the final estimate, the measurement dataset is divided into several subgroups, and then separate estimates are calculated for each subset. The median value from each subset is then used. This method improves statistical stability in the low-shot region and reduces the impact of outliers. Afterwards, the rebuilt spin-correlation matrix is assessed as

$$\hat{C}_{ij} = \frac{1}{3}\left(\langle X_i X_j \rangle + \langle Y_i Y_j \rangle + \langle Z_i Z_j \rangle\right), \tag{10}$$

where the classical shadows are used to estimate the expectation values.

Machine Learning Problem Formulation

The prediction task is formulated as a supervised regression problem, with the goal of learning a mapping from Hamiltonian parameters to the resulting correlation structure. As stated in Eq. 11, a flattened coupling vector x represents each incident of a Hamiltonian.

$$\mathbf{x} = \mathrm{vec}(\mathbf{J}) \tag{11}$$

Additionally, the flattened spin-correlation matrix is the appropriate target output, as stated in Eq. 12.

$$\mathbf{y} = \mathrm{vec}(\mathbf{C}) \tag{12}$$

The learning problem can be expressed using Eq. 13, where f_θ is the parameterised prediction model.

$$f_\theta : \mathbf{x} \to \mathbf{y} \tag{13}$$

We investigated two different types of supervision: (i) exact labels derived from ground-state diagonalisation and (ii) shadow-based labels created from classical-shadow measurements. This dual-label methodology makes it possible to systematically assess the model's performance in both ideal and measurement-noisy scenarios.

Kernel-Based Regression Models

We employ two kernel-based regression frameworks to model the nonlinear relationship between Hamiltonian parameters and the correlation structures generated from these parameters. The first method employs the radial basis function (RBF) kernel, which is based on the Gaussian distribution, and γ regulates the decay of similarity in the space of Hamiltonian parameters. The kernel ridge regression (KRR) and support vector regression (SVR) models both use this kernel.

$$K(\mathbf{x}, \mathbf{x}') = \exp\left(-\gamma \left\|\mathbf{x} - \mathbf{x}'\right\|^2\right) \tag{14}$$

In the alternative method, a multilayer perceptron is trained using fixed random parameters to produce a kernel that is induced by a neural network. The network generates an embedding $f_{\theta_0}(\mathbf{x})$ given an input $\mathbf{x}$, where θ_0 represents the weights of the network, which are initially set and randomly generated. The kernel that corresponds to this is described as

$$K(\mathbf{x}_a, \mathbf{x}_b) = \left|f_{\theta_0}(\mathbf{x}_a)\right| \cdot \left|f_{\theta_0}(\mathbf{x}_b)\right| \tag{15}$$

The construction presents a nonlinear representation of features for kernel-based learning. The suggested kernel is a fixed nonlinear embedding and differs from neural tangent kernels in that it does not engage in parameter optimisation or gradient-based linearisation.

Training Strategy and Hyperparameter Selection

The learning framework is constructed as a supervised regression problem, wherein individual elements of the spin-correlation matrix are forecasted based on Hamiltonian parameters. Each Hamiltonian instance is depicted by a flattened feature vector as specified in Eq 16, while the target variable pertains to a singular correlation element C_{ij}.

$$\mathbf{x} = \mathrm{vec}(\mathbf{J}) \tag{16}$$

This formulation allows for a thorough comparison of various kernel-based learning techniques by breaking down the prediction task into a set of scalar regression issues. We use a predetermined 70/30 split to divide the dataset into training and testing subsets. For both support vector regression (SVR) and kernel ridge regression (KRR), hyperparameter optimisation is conducted using grid search across the target set

$$\mathscr{H} = \{0.0025, 0.0125, 0.025, 0.05, 0.125, 0.25, 0.5, 1.0, 5.0, 10.0\}. \tag{17}$$

In the SVR framework, the regularisation parameter C governs the balance between model complexity and training error. The regularisation coefficient for KRR is stated by Eq. 18, which ensures a consistent comparison between the two regression methods.

$$\alpha = \frac{1}{2C} \tag{18}$$

Model selection is conducted by 5-fold cross-validation on the training dataset, utilising the root mean square error (RMSE) as the assessment metric. The hyperparameter that produces the minimum average validation error is chosen, and the associated model is retrained using the whole training dataset prior to assessment on the reserved test set. The kernel surrogate model, induced by the neural network, is assessed utilising identical training and testing partitions. The network parameters stay static upon random initialisation, and no further hyperparameter optimisation is conducted.

Unified View of Quantum Hardware and Software Ecosystems

PennyLane

PennyLane is employed in this work as the primary framework for implementing and evaluating quantum circuit simulations and expectation-value computations within a differentiable programming paradigm[26]. Its main role in the methodology is to provide a unified interface for constructing parameterized quantum circuits and evaluating observables through automatic differentiation techniques.

In the context of this study, PennyLane is used to simulate the Heisenberg ground-state observables and compute exact reference correlation matrices. Its device-agnostic architecture allows execution on both classical simulators and quantum devices, enabling consistent benchmarking across different computational regimes. Importantly, PennyLane serves as the baseline software stack for exact quantum simulation results used throughout the correlation and fidelity analysis.

Qiskit

Qiskit is utilized as the hardware-oriented quantum software framework for circuit construction, optimization, and execution on IBM quantum devices[27]. Within this study, its primary function is to translate abstract quantum circuits into hardware-compatible representations through transpilation, taking into account device connectivity, native gate sets, and noise characteristics.

Unlike PennyLane, which is used for exact simulation, Qiskit serves as the intermediate compilation layer that enables execution on real or simulated IBM hardware backends. It is used to validate the robustness of correlation estimation under realistic noise and hardware constraints, providing a bridge between theoretical simulation and physical implementation.

IBM Quantum Hardware: Circuit Implementation and Hardware-Aware Execution

The IBM quantum hardware backend is employed as the experimental evaluation platform for noisy quantum circuit execution. In this study, it is used in conjunction with a hardware-compatible simulator (IBM-FB) to evaluate expectation values of edge-based observables derived from a shallow, lattice-structured quantum circuit.

The implemented circuit does not represent the exact ground state of the Heisenberg Hamiltonian. Instead, it encodes the lattice connectivity through a shallow entangling structure composed of single-qubit Hadamard gates followed by controlled-NOT operations along the edge set of the graph. This construction allows the investigation of how hardware noise and limited connectivity affect the estimation of spin-correlation observables.

The role of the IBM hardware layer is therefore not to reproduce exact quantum physics, but to provide a realistic noisy execution environment in which the stability and consistency of the proposed correlation estimation framework can be evaluated.

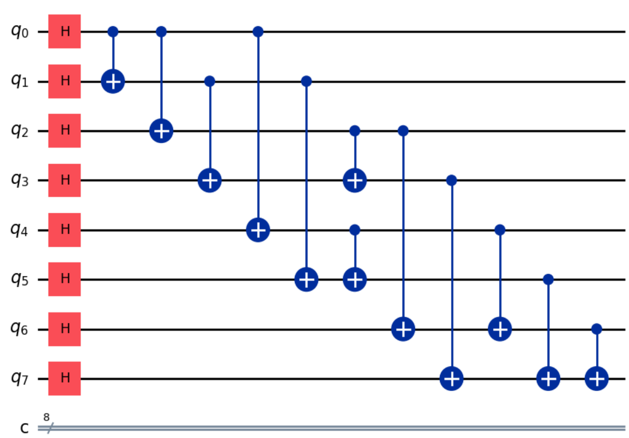

Figure 2. IBM Quantum Circuit.

Result Analysis

Experimental Design

The research employs a 3D simple cubic spin lattice with $2 \times 2 \times 2$ unit cells, resulting in $N = 8$ qubits with a lattice constant of $a = 1.0$. We construct random-bond isotropic Heisenberg Hamiltonians by uniformly sampling coupling strengths from $[0,1]$ and selecting random seeds from $[0,100]$. In PennyLane, the exact ground state for each Hamiltonian is found by sparse exact diagonalisation and is then used as a benchmark for spin-correlation calculations with the `default.qubit` backend.

Apart from that, classical shadow reconstruction is carried out with snapshot numbers $T \in \{10, 1000, 2000, 5000, 8000\}$ using single-shot Pauli measurements in order to examine the accuracy of reconstruction with distinct measurement constraints. A dataset of 100 Hamiltonian realisations is created for supervised learning, using flattened spin-correlation matrices as regression targets and flattened coupling vectors as inputs. Table 2 represent the overview of our proposed quantum-classical hybrid system architecture that is utilized in this study.

Table 2. Overview of the quantum-classical hybrid system architecture used in this study.

Component	Description
Physical system	3D cubic lattice with size $2 \times 2 \times 2$ corresponding to $N = 8$ qubits.
Interaction model	Random-bond isotropic Heisenberg Hamiltonian with uniform couplings $J_{ij} \sim \mathscr{U}(0,1)$.
Hamiltonian connectivity	Nearest-neighbour lattice graph with 12 edges.
Ground-state computation	Exact diagonalization using sparse eigensolver; ground state $\|\psi_0\rangle$ used as reference.
Quantum observable	Isotropic spin correlation $C_{ij} = \frac{1}{3}(\langle X_i X_j \rangle + \langle Y_i Y_j \rangle + \langle Z_i Z_j \rangle)$.
Exact simulation framework	PennyLane statevector simulation used for benchmark expectation values.
Measurement model	Classical shadow protocol with random Pauli measurements and snapshot size $T \in \{10, 1000, 2000, 5000, 8000\}$.
Machine learning task	Regression mapping from Hamiltonian parameters $\mathrm{vec}(J)$ to correlation matrix entries $\mathrm{vec}(C)$.
Learning models	Gaussian Kernel Regression (GKR) and Neural-network-induced kernel surrogate (NNIKS).
Training protocol	70/30 train-test split with 5-fold cross-validation for hyperparameter selection.
Hardware validation	IBM-FakeFez noisy backend for circuit execution and correlation estimation.
Execution parameters	5000 shots per circuit evaluation with resilience level set to 1.

Evaluation Metrics

The performance of the proposed framework is evaluated using three complementary metrics that jointly quantify quantum state reconstruction quality, observable estimation accuracy, and ML predictive performance. These metrics are defined with respect to a system of $N = N_x N_y N_z$ qubits, where the exact ground state is denoted by $|\psi_0\rangle$, the reconstructed density matrix obtained from the CS protocol is denoted by $\hat{\rho}$, the exact spin-correlation matrix computed from ground-state expectation values is denoted by $C \in \mathbb{R}^{N \times N}$, and the corresponding estimated correlation matrix obtained either from CS or ML predictions is denoted by $\hat{C} \in \mathbb{R}^{N \times N}$.

Fidelity

The similarity between the reconstructed quantum state and the exact ground state is quantified using the fidelity measure, defined as

$$F = \langle \psi_0 | \hat{\rho} | \psi_0 \rangle \tag{19}$$

where $|\psi_0\rangle$ is the exact ground-state wavefunction obtained from sparse diagonalization of the Heisenberg Hamiltonian, and $\hat{\rho}$ represents the density matrix reconstructed using the classical shadow protocol with T measurement snapshots. The fidelity value lies in the range $F \in [0,1]$, where $F = 1$ corresponds to a perfect reconstruction of the quantum state, while lower values indicate deviations introduced by finite sampling and measurement noise.

Root Mean Square Error (RMSE)

The accuracy of reconstructed or predicted spin-correlation matrices is evaluated using the root mean square error between the exact correlation matrix C and its $\hat{C}$. The RMSE is defined as

$$\mathrm{RMSE}(C,\hat{C}) = \sqrt{\frac{1}{N^2}\sum_{i=1}^{N}\sum_{j=1}^{N}\left(C_{ij}-\hat{C}_{ij}\right)^2} \tag{20}$$

Here, C_{ij} denotes the exact isotropic spin correlation computed from the ground state $|\psi_0\rangle$, while $\hat{C}_{ij}$ represents either (i) the correlation matrix reconstructed using classical shadow measurements or (ii) the predicted correlation matrix obtained from kernel-based regression models. The normalization factor $1/N^2$ ensures size-independent comparison across different lattice dimensions.

Maximum Absolute Deviation

To capture worst-case reconstruction errors that may not be reflected in average performance metrics, the maximum absolute deviation is defined as

$$\Delta_{\max}(C,\hat{C}) = \max_{1\le i,j\le N}\left|C_{ij}-\hat{C}_{ij}\right| \tag{21}$$

where C_{ij} and $\hat{C}_{ij}$ are defined as above. This metric highlights the largest element-wise discrepancy between exact and estimated correlation matrices and is particularly sensitive to localized reconstruction failures arising from measurement noise, finite-shot effects, or model misprediction.

Cross-Validation and Test RMSE

The machine learning component of the framework is evaluated using both cross-validation RMSE and test RMSE. In this setting, each Hamiltonian realization is encoded as a feature vector $x = \mathrm{vec}(J)$, while each correlation entry is treated as a regression target $\hat{C}_{ij}$. Cross-validation RMSE is computed over five independent folds of the training dataset to assess generalization performance during model selection, while test RMSE is computed on a held-out dataset that is not used during training or hyperparameter tuning.

For each kernel-based regression model, including Gaussian kernel regression and neural-network-induced kernel regression, these metrics provide a consistent evaluation of predictive accuracy. The cross-validation RMSE is used for hyperparameter selection, while the test RMSE is used for final performance reporting. Together, these metrics ensure a rigorous and unbiased assessment of model generalization in predicting quantum correlation structures from Hamiltonian parameters.

Element-wise Comparative Analysis of Correlation Values: PennyLane, Classical Shadow, Qiskit AER, and IBM-FB hardware.

Table 3 presents the element-wise correlation values generated by PennyLane, CS, Qiskit, and IBM-FB hardware. The prevailing trend indicates that CS continually aligns closely with the PennyLane baseline across the majority of entries, maintaining both the sign structure and the approximate magnitude of the correlations. This is evident in representative entries such as C_{01}, C_{14}, and C_{57}, where the shadow estimates differ only slightly from the exact values. The practical implication is that the shadow protocol successfully preserves the physically relevant two-body structure of the many-body ground state, despite relying on randomized measurements rather than direct access to the exact state.

Qiskit replicates the overall correlation structure, albeit with some localized aberrations that exceed those reported for CS. In particular, entries such as C_{02}, C_{03}, and C_{63} depart more noticeably from the PennyLane baseline, indicating that the Qiskit results should be interpreted as *broadly consistent* rather than numerically identical to the exact reference. This divergence is crucial for methodological rigor: the Qiskit branch facilitates cross-framework validation of the correlation pattern; however, it does not confirm exact equivalence at the entrywise level.

The IBM-FB outcomes exhibit the most significant divergence from the simulation-based benchmarks. Numerous off-diagonal entries are diminished or rendered null, and various negative correlations in PennyLane, CS, and Qiskit either weaken significantly or vanish completely. For example, C_{03} changes sign relative to the simulation-based results, while strongly negative entries such as C_{57} and C_{75} vanish in the hardware matrix. This behavior is consistent with the experimental design: the IBM-FB branch does not prepare the Heisenberg ground state, but instead evaluates edge-wise observables on a proxy graph-entangling circuit. Consequently, the hardware matrix should be regarded as a genuine device benchmark for proxy-circuit observables rather than as a direct manifestation of the precise many-body state.

The matrix visualizations in Figures 3–6 reinforce the same conclusion. Figure 3 shows that Qiskit and PennyLane share the same broad correlation structure, while the corresponding difference map confirms that the mismatch is concentrated in a limited subset of off-diagonal entries. Figures 4 and 5 show a qualitatively different regime for IBM-FB hardware: the hardware-derived matrix exhibits more sparsity and reduced strength, with the difference maps predominantly characterized by a limited number of significant discrepancies rather than several uniformly distributed minor mistakes. Figure 6 illustrates the

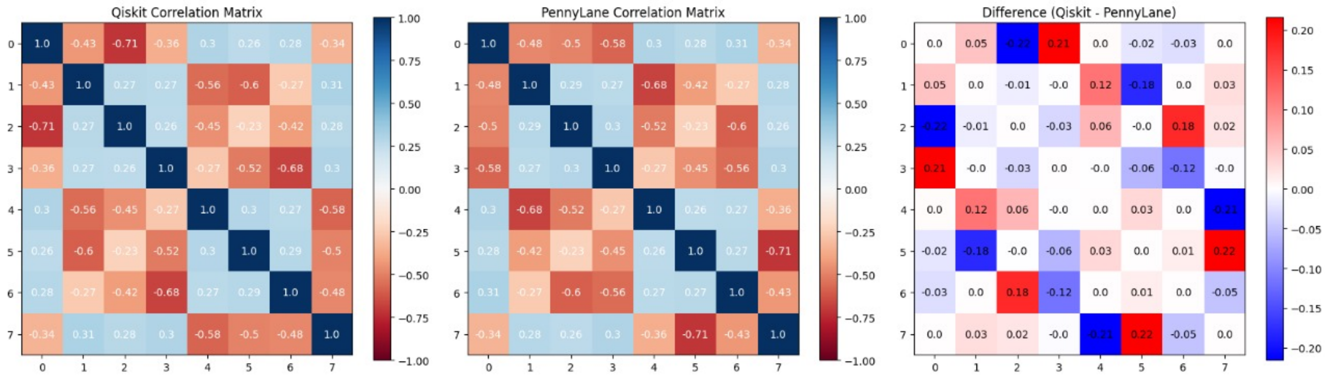

Figure 3. Correlation matrix comparison: Qiskit vs PennyLane.

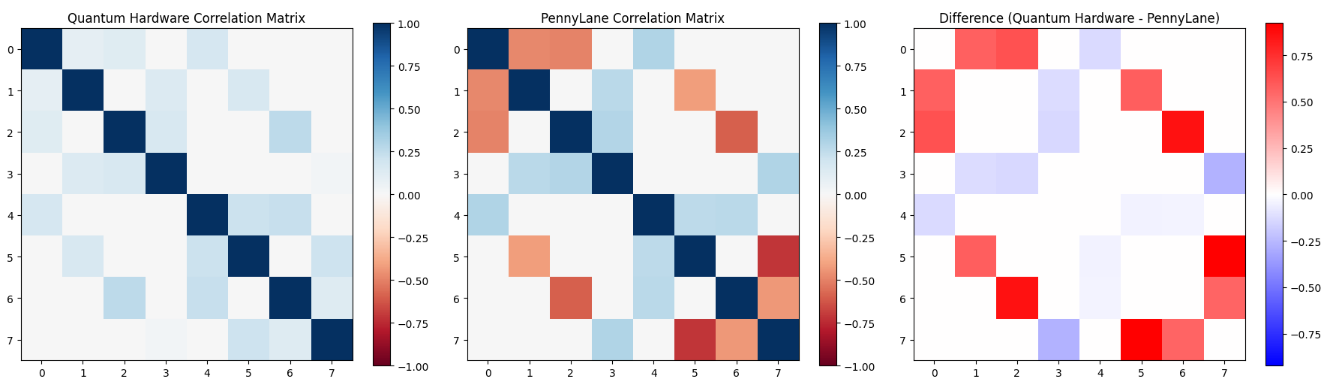

Figure 4. Correlation matrix comparison: IBM-FB hardware vs PennyLane.

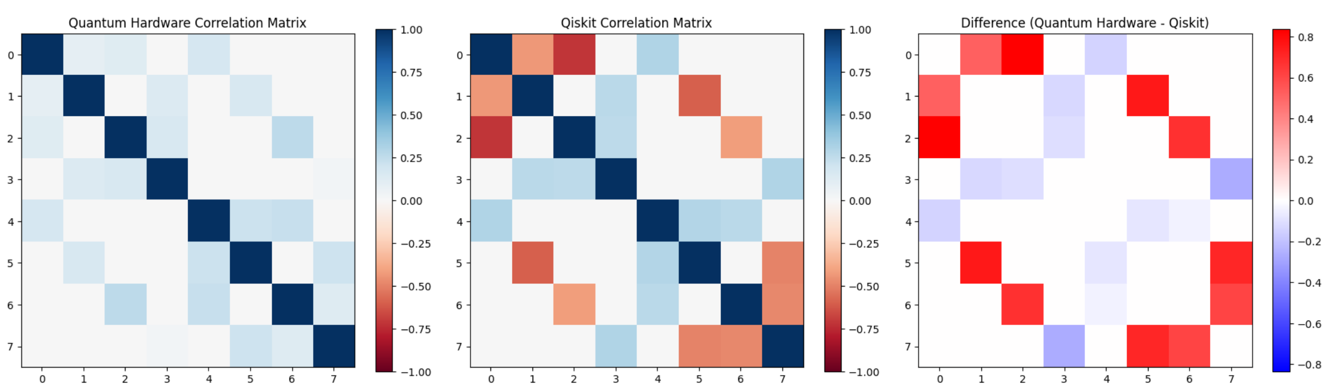

Figure 5. Correlation matrix comparison: IBM-FB hardware vs Qiskit.

Table 3. Element-wise correlation values from PennyLane (PL), Classical Shadows (CS), Qiskit, and IBM-FakeFez Backend (IBM-FB).

Pair	PL	CS	Qiskit	IBM-FB	Pair	PL	CS	Qiskit	IBM-FB
C00	1.0000	1.0000	1.0000	0.00	C40	0.3007	0.2750	0.3030	0.00
C01	-0.4798	-0.4815	-0.4308	0.09028	C41	-0.6783	-0.6724	-0.5618	0.00
C02	-0.4983	-0.4540	-0.7139	0.12463	C42	-0.5166	-0.4277	-0.4548	0.00
C03	-0.5783	-0.5287	-0.3642	0.16005	C43	-0.2715	-0.2374	-0.2715	0.00
C04	0.3007	0.2750	0.3030	0.00	C44	1.0000	1.0000	1.0000	0.00
C05	0.2803	0.2641	0.2591	0.00	C45	0.2639	0.1890	0.2965	0.13060
C06	0.3133	0.3076	0.2848	0.00	C46	0.2661	0.2714	0.2669	0.00
C07	-0.3379	-0.2906	-0.3379	0.00	C47	-0.3642	-0.3056	-0.5783	0.00
C10	-0.4798	-0.4815	-0.4308	0.00	C50	0.2803	0.2641	0.2591	0.00
C11	1.0000	1.0000	1.0000	0.00	C51	-0.4159	-0.4758	-0.5951	0.00
C12	0.2878	0.3333	0.2738	0.13987	C52	-0.2334	-0.2643	-0.2334	0.00
C13	0.2669	0.1576	0.2661	0.15898	C53	-0.4548	-0.5238	-0.5166	0.00
C14	-0.6783	-0.6724	-0.5618	0.13053	C54	0.2639	0.1890	0.2965	0.00
C15	-0.4159	-0.4758	-0.5951	0.00	C55	1.0000	1.0000	1.0000	0.00
C16	-0.2656	-0.2374	-0.2656	0.00	C56	0.2738	0.3524	0.2878	0.00
C17	0.2848	0.2708	0.3133	0.00	C57	-0.7139	-0.7380	-0.4983	0.00
C20	-0.4983	-0.4540	-0.7139	0.00	C60	0.3133	0.3076	0.2848	0.00
C21	0.2878	0.3333	0.2738	0.00	C61	-0.2656	-0.2374	-0.2656	0.00
C22	1.0000	1.0000	1.0000	0.00	C62	-0.5951	-0.5859	-0.4159	0.00
C23	0.2965	0.3449	0.2639	0.25813	C63	-0.5618	-0.6185	-0.6783	0.00
C24	-0.5166	-0.4277	-0.4548	0.02614	C64	0.2661	0.2714	0.2669	0.00
C25	-0.2334	-0.2643	-0.2334	0.21065	C65	0.2738	0.3524	0.2878	0.00
C26	-0.5951	-0.5859	-0.4159	0.00	C66	1.0000	1.0000	1.0000	0.00
C27	0.2591	0.3060	0.2803	0.00	C67	-0.4308	-0.4530	-0.4798	0.00
C30	-0.5783	-0.5287	-0.3642	0.00	C70	-0.3379	-0.2906	-0.3379	0.00
C31	0.2669	0.1576	0.2661	0.00	C71	0.2848	0.2708	0.3133	0.00
C32	0.2965	0.3449	0.2639	0.00	C72	0.2591	0.3060	0.2803	0.00
C33	1.0000	1.0000	1.0000	0.00	C73	0.3030	0.1254	0.3007	0.00
C34	-0.2715	-0.2374	-0.2715	0.22343	C74	-0.3642	-0.3056	-0.5783	0.00
C35	-0.4548	-0.5238	-0.5166	0.20618	C75	-0.7139	-0.7380	-0.4983	0.00
C36	-0.5618	-0.6185	-0.6783	0.00	C76	-0.4308	-0.4530	-0.4798	0.00
C37	0.3030	0.1254	0.3007	0.00	C77	1.0000	1.0000	1.0000	0.00

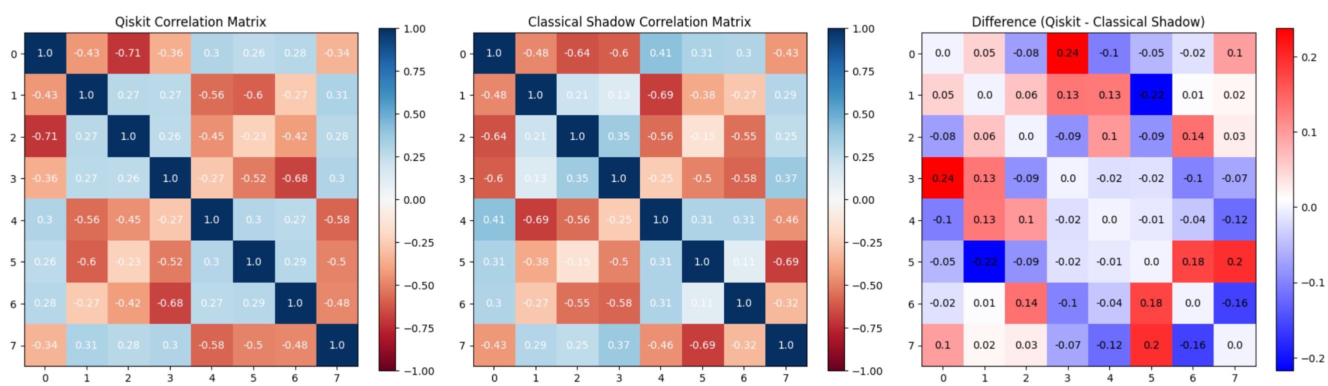

Figure 6. Correlation matrix comparison: Qiskit vs CS.

correlation matrix comparison between Qiskit and CS. This pattern aligns with the experimental configuration, wherein only nearest-neighbor edge observables are assessed on a proxy circuit, while the other off-diagonal entries are inherently zero.

Cross-Platform Analysis of Correlation Values

The element-wise observations from Table 3 are quantitatively consolidated by the maximum-deviation and RMSE results reported in Tables 4 and 5.

Table 4 shows the absolute deviations between correlation matrices computed using various computational platforms. There is significant alignment and confirmation of the proposed computational framework's consistency, indicated by the modest variances across the simulation-based approaches, specifically PennyLane versus CS (0.16267), PennyLane vs Qiskit (0.21555), and Qiskit vs CS (0.23796). The larger deviations observed in comparisons between IBM-FB and CS (0.7280), Qiskit vs IBM-FB (0.83629), and PennyLane vs IBM-FB (0.92307) are anticipated as a result of noise, finite-shot effects, gate errors, and hardware calibration limitations. Despite these hardware-induced evolutions, the results are promising because the ideal and shadow-based platforms remained nearly identical, and the IBM-FB results represent an acceptable baseline for assessing the performance under noisy quantum hardware conditions.

Table 4. Maximum absolute deviations across computational platforms for correlation matrices.

Comparison	Maximum Difference
PennyLane vs Classical Shadow	0.16267
PennyLane vs Qiskit	0.21555
PennyLane vs IBM-FB	0.92307
Qiskit vs Classical Shadow	0.23796
Qiskit vs IBM-FB	0.83629
IBM-FB vs Classical Shadow	0.7280

Table 5. RMSE for correlation matrices across different platforms.

Comparison	RMSE
PennyLane vs Classical Shadow	0.06604
PennyLane vs Qiskit	0.09575
PennyLane vs IBM-FB	0.11992
Qiskit vs Classical Shadow	0.10398
Qiskit vs IBM-FB	0.45377
IBM-FB vs Classical Shadow	0.3814

Table 5 provides a thorough evaluation of the comprehensive consistency at the matrix level by exhibiting the RMSE values calculated between correlation matrices produced across various platforms. The shadow-based estimation approach and ideal simulation are consistent, with PennyLane vs Classical Shadow having the lowest RMSE (0.06604). Additionally, PennyLane vs. Qiskit (0.09575) and Qiskit vs. Classical Shadow (0.10398) provide minimal RMSE values, which strengthen the significant level of correlation between the simulation and estimation-based methods. Furthermore, the comparison between PennyLane and IBM-FB reveals a moderate RMSE of 0.11992, indicating that the findings obtained from IBM-FB hardware continue to demonstrate a significant convergence with the ideal reference even after accounting for actual noise effects. The influence of hardware-related imperfections, including gate noise, readout errors, finite-shot variation, and calibration drift, is evident in the higher RMSE values observed for IBM-FB vs CS (0.3814) and Qiskit vs IBM-FB (0.45377). In a broader sense, these findings suggest that the proposed framework obtains consistent performance across multiple platforms, while the IBM-FB results offer valuable real-world validation under the challenging conditions of quantum noise.

The outcomes prove that the suggested hybrid quantum-classical model is legitimate by showing that the PennyLane, Qiskit, and CS estimations are highly consistent at the matrix level. Also, results from IBM-FB reveal expected deviations due to noisy intermediate-scale quantum technology, but the observed consistency proves the numerical stability and reliability of the correlation-reconstruction pipeline. In general, the framework offers a technically sound and hardware-aware method for the analysis of many-body ground-state correlations.

Figure 7 represents the PennyLane correlation matrix, and Figure 8 demonstrates a clear depiction of the error structure associated with the CS and PennyLane. The error map predominantly exhibits minor residuals, aligning with the low RMSE and moderate maximum deviation; yet, a few off-diagonal entries are significantly bigger than the background level. Consequently, the correct conclusion is that the shadow reconstruction effectively represents the primary correlation structure with substantial overall fidelity, but with finite-sample statistical errors in a limited range of observables.

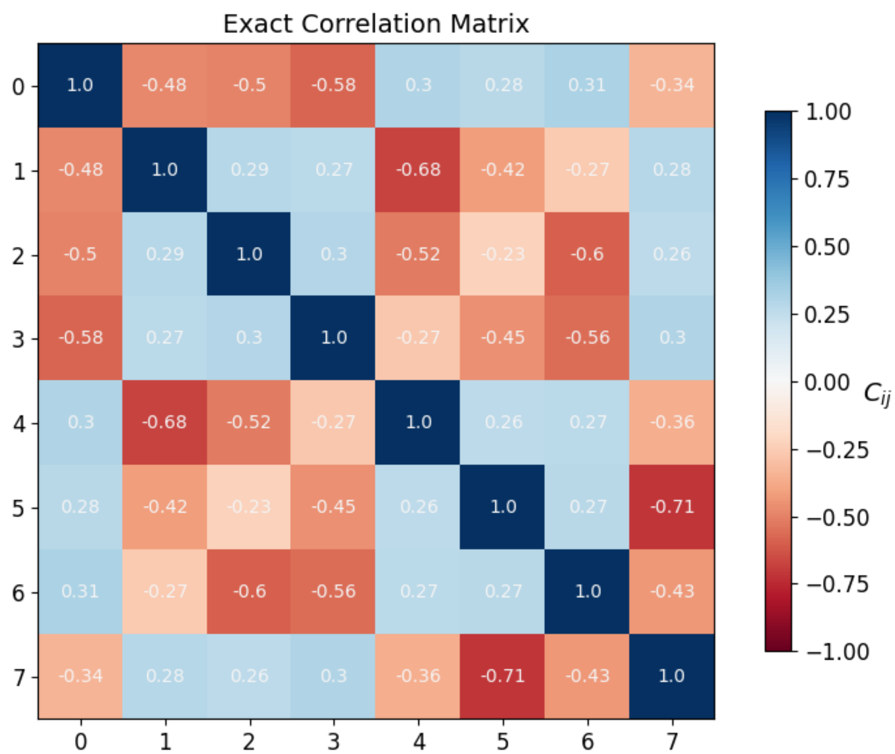

Figure 7. PennyLane correlation matrix.

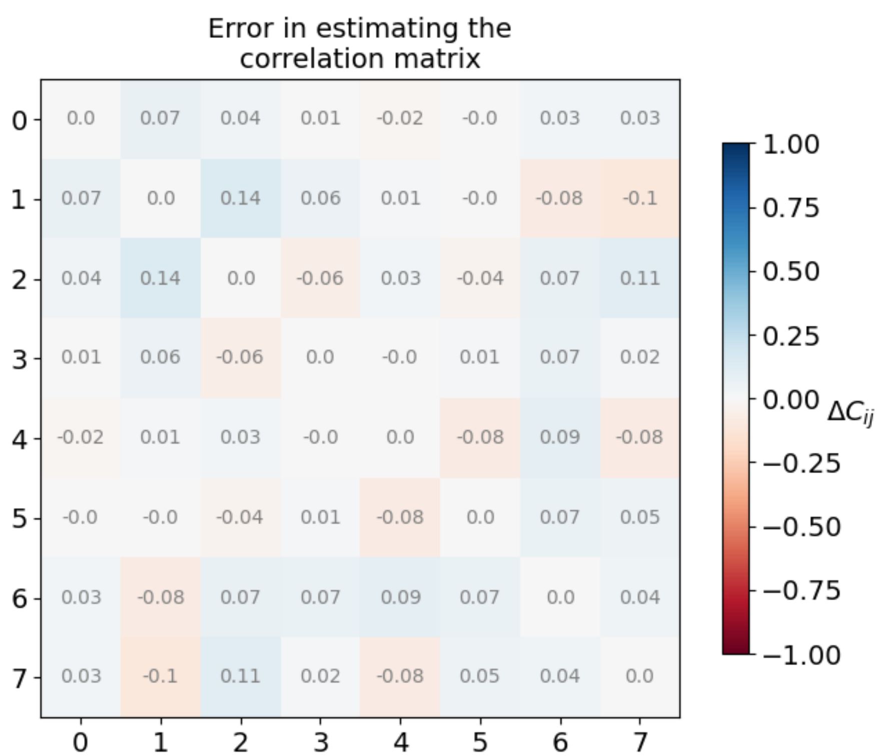

Figure 8. Heatmap visualization of the exact PennyLane correlation matrix and the element-wise estimation error obtained using the CS method.

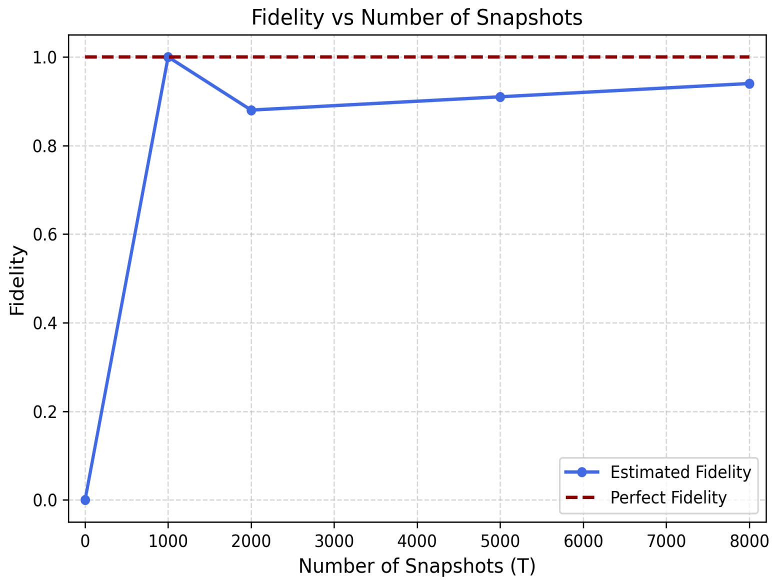

Figure 9. Fidelity of CS reconstructed states versus the number of measurement snapshots.

Figure 9 shows that as the number of snapshots T rises, the fidelity performance improves. This highlights the efficacy of the suggested method with a small number of snapshots, as the estimated fidelity quickly approaches the ideal value at T=1000. However, the fidelity remains consistently high and progressively improves as T increases, reaching approximately 0.94 at T=8000, despite a slight variation at intermediate snapshot numbers. The results show that the method is stable and dependable for high-fidelity estimates, which confirms its robustness and potential scalability.

Kernel-Based Learning Performance

Table 6 extends the analysis from correlation estimation to correlation prediction. The GKR model achieves the lowest error in both validation and held-out test settings, with RMSE values of 0.06141 and 0.11337, respectively, whereas the NNIKS yields higher errors of 0.09831 and 0.15420. The identified performance disparity is consistent across both evaluation phases and suggests that the GKR offers a more appropriate inductive bias for learning the coupling-to-correlation mapping in the examined finite-size Heisenberg system.

Table 6. Machine-learning performance: cross-validation and test RMSE for GKR and NNIKS models.

Kernel Model	RMSE(CV Score)	RMSE(Test Score)	Interpretation
NNIKS	0.09831	0.15420	Higher validation and test errors
GKR	0.06141	0.11337	Best overall predictive performance

Also, this study's ML contribution should not be viewed solely as a generic predictive extension, but as evidence that the many-body correlation map can be directly learned from microscopic coupling parameters, given that the training targets are produced through a physically grounded quantum framework. The results in Table 6 therefore demonstrate that the proposed approach enables not only exact benchmarking and measurement-efficient reconstruction, but also reliable data-driven prediction of correlation structures, with the GKR emerging as the most effective model in the present setting.

The aggregate interpretation of the results is summarized in Table 7. PennyLane functions as the definitive reference baseline, whilst CS offers the nearest practical approximation. Qiskit provides a significant cross-platform consistency assessment without requiring precise numerical equivalency, while IBM-FB emphasizes the disparity between idealized simulation and real device execution. During the supervised learning phase, the GKR is identified as the most efficacious prediction model.

An 8-qubit 3D Heisenberg spin lattice's quantum circuit is illustrated in Figure 10. The lattice interactions are encoded by Hadamard and CNOT gates, and correlation measurements in the X, Y, and Z bases are enabled via parameterised $\sqrt{X}$ and R_Z rotations. In hardware experiments and simulations, this circuit is utilised for direct correlation estimation and CS reconstruction.

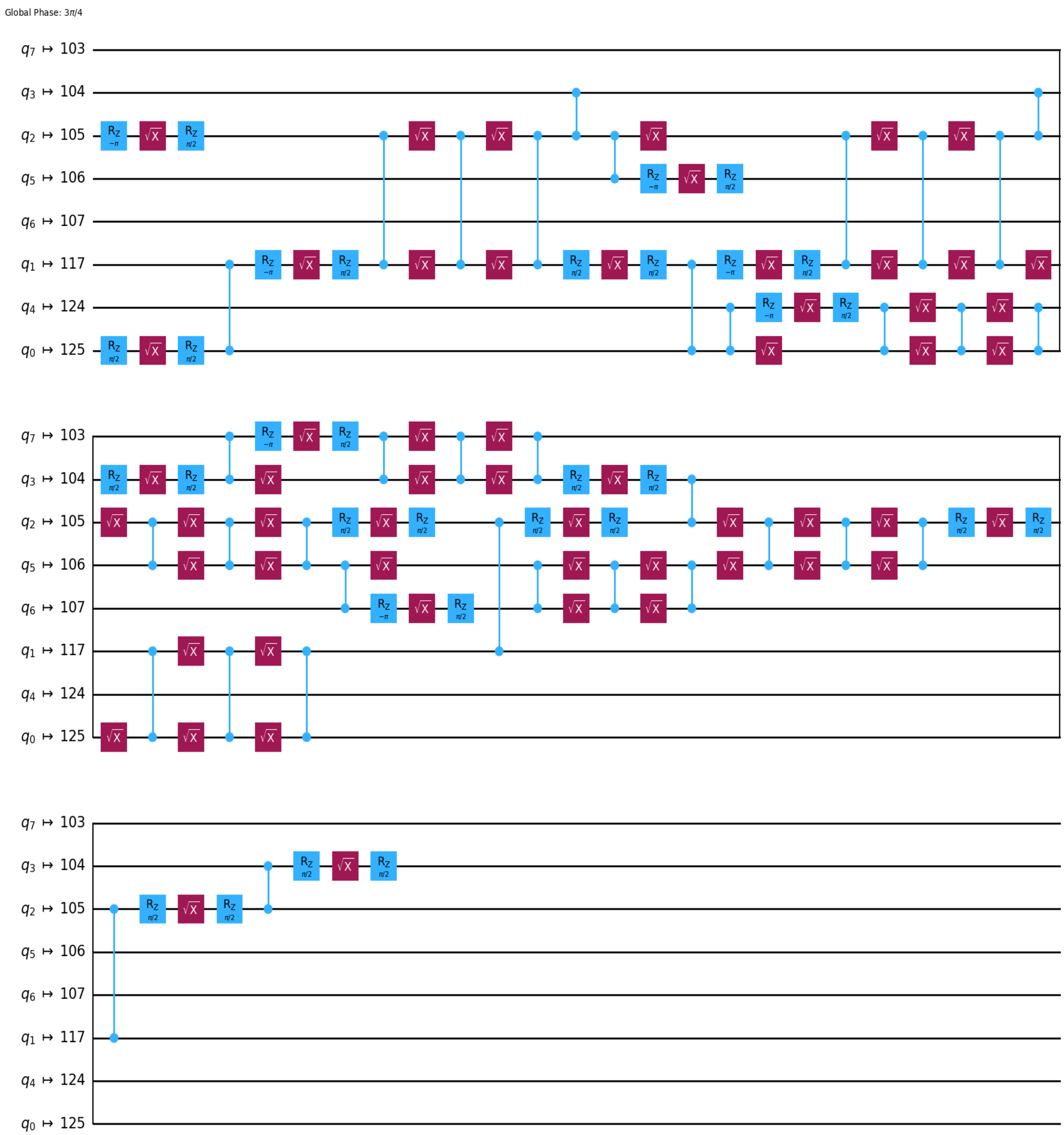

Figure 10. ISA-transpiled 8-qubit quantum circuit mapped to the IBM-FB for implementing nearest-neighbor interactions in the 3D Heisenberg lattice model.

Table 7. Overview of the research findings.

Aspect	Main Finding
Exact reference baseline	PennyLane
Shadow approximation	CS showed the closest agreement with PennyLane
Cross-framework validation	Qiskit was broadly consistent with PennyLane
Hardware comparison	IBM-FB showed the largest deviation from simulation-based results
Best ML kernel	GKR

Table 8. List of abbreviations.

Abbreviation	Full Form	Abbreviation	Full Form
CS	Classical Shadow	QAOA	Quantum Approximate Optimization Algorithm
PL	PennyLane	QGAN	Quantum Generative Adversarial Network
RMSE	Root Mean Square Error	QCNN	Quantum Convolutional Neural Network
GKR	Gaussian-kernel Regression	GNN	Graph Neural Network
NNIKS	Neural-Network-Induced Kernel Surrogate	CV	Cross Validation
ML	Machine Learning	VQA	Variational Quantum Algorithms
QML	Quantum Machine Learning	SGD	Stochastic Gradient Descent
CST	Classical Shadow Tomography	MLP	Multi-Layer Perceptron
NISQ	Noisy Intermediate-Scale Quantum	SVR	Support Vector Regression
RL	Reinforcement Learning	RBF	Radial Basis Function
CNN	Convolutional Neural Network	KRR	Kernel Ridge Regression
RShadow	Robust Shadow Estimation	NTK	Neural Tangent Kernels
RBM	Restricted Boltzmann Machine	2D	Two Dimensional
MPS	Matrix Product State	3D	Three Dimensional
DNN	Deep Neural Network	FFT	Fermionic Fourier Transformation
VQE	Variational Quantum Eigensolver	ISA	Instruction-Set-Architecture
QNN	Quantum Neural Network	IBM-FB	IBM FakeFez Backend

Conclusions and Future Work

This paper presents a cohesive hybrid quantum-classical framework for analyzing finite many-body correlations in a $2 \times 2 \times 2$ random-bond isotropic Heisenberg lattice. The suggested framework methodically incorporates the following modules: PennyLane for sparse exact diagonalisation, CS for observable reconstruction, Qiskit for cross-platform validation, IBM-FB for hardware-aware noisy backend benchmarking, and kernel-based prediction from microscopic Hamiltonian couplings. The architecture creates a coherent connection between data-driven surrogate modelling, measurement-efficient estimation, exact numerical benchmarking, and hardware-aware noisy assessment through this integrated design. According to the findings, Qiskit has strong consistency at the software-validation level, and CS reconstruction gives the best practical approximation to the exact PennyLane benchmark. However, the disparities seen with the IBM-FB illustrate the anticipated impact of proxy-state mismatch, finite-shot sampling effects, backend-informed noise, and limitations imposed by hardware-aware transpilation. Also, at the predictive stage, the NNIKS is beaten by GKR, suggesting that the coupling-to-correlation connection in the current finite lattice is well-structured enough for surrogate prediction to be reliable. The findings show that the suggested framework for finite many-body quantum correlation analysis is robust, scalable, and hardware-aware.

The current investigation has certain limitations, such as an 8-qubit lattice that is finite, a dataset consisting of one hundred Hamiltonian realisations, and the substitution of a surrogate IBM-FB hardware-aware noisy circuit for a completely confirmed Heisenberg ground-state preparation technique. These constraints specify the results' scope without diminishing the suggested framework's contribution. In the future, the approach should be expanded to encompass a wider range of lattice structures, integrate validated hardware-state preparation with error mitigation, and employ more sophisticated physics-informed or graph-based learning models. All of these enhancements would enhance the framework's practical applicability, reliability, and scalability in the context of realistic quantum many-body systems.

APPENDIX

For the readers' convenience, Table 8 enumerates the abbreviations often employed in this research work.

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